PV101 Day 06 Instructor Run Sheet

Tuesday, July 14, 2026 | CTS Cohort 6 | Troubleshooting and Hidden Fuses

Learning Targets

- Trace power flow through a small off-grid PV system.
- Use voltage and continuity measurements to isolate an open circuit.
- Explain how a fuse protects conductors and equipment.
- Troubleshoot before replacing parts.
- Document evidence clearly.

Timeline

Time	Block	Instructor Move
9:00-9:15	Opening	Frame the day: the smallest cheapest part can stop the whole system.
9:15-9:35	Quiz	10 questions on fuses, meters, continuity, and power flow.
9:35-10:15	Teach	Power-flow review, fuse purpose, and the verify-before-repair mindset.
10:15-10:45	Demo	Show voltage tracing without revealing today's fault.
10:45-11:30	Practice	Students mark likely test points on a wiring diagram.
12:30-2:15	Lab	Hidden Fuse Challenge in three teams.
2:15-2:45	Debrief	Each team explains what they checked and what proved the failure.
2:45-3:00	Reflection	Students write how today's process changes their troubleshooting.

Safety Brief

- Stop and de-energize before opening fuse holders.
- Measure voltage before touching exposed conductors.
- Never bypass a fuse.
- Use the correct meter setting before each measurement.

Whiteboard

STOP -> LOOK -> THINK -> MEASURE -> VERIFY -> THEN REPAIR